

Mingli Zhang

☎ (919)274-5562 | ✉ minglizhang2000@outlook.com | 🏠 www.minglizhang.com | 📷 /MingliZhang | 🌐 /in/mingli-zhang/

Objective

Motivated and skilled professional seeking a role as a Machine Learning Engineer or Full-Stack Developer. Offering hands-on experience in developing AI-driven solutions and web/mobile applications to create impactful, user-focused products. Dedicated to continuous learning, contributing to collaborative teams, and evolving into a mentorship role to support and guide emerging talent in the tech industry.

Work Experience

ByteDance | Machine Learning Backend Engineer | San Jose, CA, USA

Jan 2024 - Present

IT Enterprise Solution AI Operations Algorithm Team

- Built a Text-to-SQL service using **LangChain** and **RAG** to automate natural language-to-SQL queries, which enhanced our own network monitoring service, saving **100+ human hours monthly** across all offices.
- Developed and fine-tuned **computer vision** models such as **YOLO**, **DeepSort**, and **Mmpo**s to automate server room monitoring and alert service across **1,000+** global sites, saving **2000+** man hours required each year.
- Implemented a network RCA AI chatbot for a commercialized product based on **Swarm**, enabling non-technical users to diagnose network issues, which **halved the workload** of oncall technicians.
- Developed backend services using **FastAPI** and **gRPC** to expose machine learning models and support client interaction. Implemented **authentication**, **managed deployment**, and ensured service health. Led real-time output **streaming** to enhance user experience and provide instant feedback.
- Patented an algorithmic approach and co-owned 3 other patents involving machine learning solutions for real-world problems.
- Led tech shares to present lessons learned from each project, including models, fine tuning, agent frameworks, and non-ML solutions, which fostered team knowledge and improved efficiency.
- Collaborate with clients, product managers, and team members to clarify requirements, manage expectations, and deliver machine learning solutions on time through clear communication.

Cymantix | Software Engineer (Part Time) | North Carolina, USA

Oct 2021 - Feb 2023

A SaaS tech startup that integrated machine learning with data analytics and information retrieval.

- Identified and resolved **communication issues** proactively between clients and engineers, which boosted the company's reputation and gained 3 additional clients.
- Investigated replacing K-Means with Hierarchical clustering on the **Flask (Python)** backend, which reduced computation and user wait time by 10%-60%.
- Optimized real-time news analysis using **multithreading**, **NodeJS** and **Socket.io**, which decreased user wait time by 90%.
- Created a full-stack machine-learning-driven data analytics web platform in a global team of 5 engineers that secured seed-round funding.
- Refactored codebase to **Next.js** and added animations using **D3.js**, which reduced 20% code redundancy and provided a smoother user experience.
- Integrated **Microsoft Azure** into our platform, which provided login options and secure data storage from Microsoft's cloud. Resulted in a cost-effective transition for users and gained 2 non-profit organizations as clients within 1 week.
- Automated PubMed data update pipeline using **Bash script**, **Python**, and **CronJob**, which reduced manual labor and gained 2 clients that required the most up to date data.

IBM | Full-Stack Engineer (Summer Intern) | North Carolina, USA

May 2021 - Aug 2021

Worked on IBM's internal platform that provisions and customizes live demo environments.

- Co-founded an automated validation service using **SailsJS** that streamlined database maintenance and saved the team \$14,000 annually by eliminating headcounts for manual tasks.
- Enhanced user experience based on user feedback using **NextJS**, which expanded the monthly user base to over 12,000+.
- Streamlined administrative content processes, achieved faster content updates, and reduced redeployments by over 20 per week.

Education

University of North Carolina at Chapel Hill

Master's of Science in Computer Science | Aug 2022 - Aug 2023

- GPA: 3.9/4.0.

Bachelor's of Science in Computer Science and Mathematics | Aug 2018 - May 2022

- GPA: 3.8/4.0, Dean's List from 2018-2022.

Skills

Coding	JavaScript/TypeScript, SQLite, Python(Pandas, TensorFlow, NumPy, Scikit-learn, etc.), C/C++, HTML/CSS, Bash, Java
Technologies	LangChain, Swarm, NextJS/React, NodeJS, React Native, Git, Flask, FastAPI, JQuery, Microsoft Azure, Firebase, Linux, Tableau, MySQL
Hobbies	Cooking, Barista, and Tech